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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 03/11/2008 |
| **PI Name:** | Hargens, Alan R. Ph.D. | | |
| **Project Title:** | LBNP Treadmill and Resistive Exercises during Long-term Bed Rest | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Physiology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 92037-0863 | **Congressional District:** | 52 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 03-OBPR-05 |
| **Start Date:** | 07/01/2004 | **End Date:** | 09/30/2007 |
| **No. of Post Docs:** | 3 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 5 |
| **No. of Master's Candidates:** | 5 | **No. of Bachelor's Degrees:** | 4 |
| **No. of Bachelor's Candidates:** | 4 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 9/30/2007 per PI (8/07) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Schneider, Suzanne Ph.D. ( University of New Mexico )<br>Macias, Brandon ( UCSD )<br>Smith, Scott Ph.D. ( NASA Johnson Space Center )<br>Watenpaugh, Donald Ph.D. ( Sleep Consultants, Inc. )<br>Lee, Stuart M.S. ( Wyle Laboratories )<br>Levine, Ben M.D. ( UT Southwestern )<br>Guinet, Patrick M.D. ( Univ of Lyon )<br>Boda, Wanda Ph.D. ( Sonoma State University ) | | |
| **Grant/Contract No.:** | NNJ04HF71G | | |
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| **Performance Goal Text:** | | | |

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| **Task Description:** | Presently, exercise countermeasures and equipment for long-term space flight are unresolved. Calculations suggest that all exercise in space to date has lacked sufficient loads to maintain preflight musculoskeletal and cardiovascular mass and function. Previously we found that lower body negative pressure (LBNP) produces a comfortable footward "gravitational" force equal to or greater than that on Earth. Simultaneously, we have successfully generated metabolic and cardiovascular loads which equal those present in upright exercise. Thus, this artificial gravity concept provides an integrated countermeasure and has maintained physiologic systems of bed rested volunteers for up to 30 days of simulated microgravity. Currently-available bungee cord assisted, treadmill exercise is limited by harness discomfort, lower than normal loads, abnormal postflight gait, and the absence of gravitational blood pressures within the vascular system. Resistive exercise alone does not provide gravitational blood pressures and was relatively ineffective during a 90-day bed rest study of males in 2001-2002.<br>We hypothesize that our treadmill exercise protocol within LBNP, followed immediately by a short period of resting LBNP, and supplemented by high-resistance exercise will maintain leg muscle strength, orthostatic tolerance, plasma volume, upright exercise capacity, submaximal exercise responses, bone homeostasis (by reducing osteoclastic activity and increasing osteoblastic activity), paraspinal muscle cross-sectional area, and other physiologic parameters during microgravity simulated by 60 days of head-down tilt (HDT) bed rest. Our exercise countermeasure has been highly successful in studies of identical twins as bedrest volunteers with one twin having 40 minutes dynamic LBNP plus 5 minutes static LBNP, while their brother/sister acts as a non-exercise control. This experimental design reduced genetic variability and documented efficacy in previous 30-day HDT studies of males and females. Our exercise times are less than half those currently prescribed for long-duration flight. Furthermore, our exercise concept allows comfortable loads greater than one body weight and blood pressure gradients similar to those present in upright posture on Earth.<br><br>To document further the efficacy of the our exercise concept, we will continue our studies for longer durations of bed rest and include resistive exercise for improved bone and muscle maintenance. Previously, LBNP treadmill exercise was found to suppress osteoclast activity, but it failed to stimulate osteoblast activity. Differences between males and females in terms of their responses to bed rest and exercise will continue to be explored in the proposed 60 day study. The long-term goal is to provide an effective, early, safe, and economical form of artificial gravity for Space Station and Exploration missions with prolonged exposure of crew members to microgravity.<br><br>The long-term objective of this research is to test load-bearing exercise within a LBNP chamber along with high intensity resistance exercise to maintain physiologic systems during long-term exposure to simulated microgravity. We believe development of this integrated countermeasure will provide: 1) a more physiologic level and type of biomechanical stress than previously-proposed countermeasures for long-duration space flight, 2) sufficient and comfortable loading to maintain structure and function of musculoskeletal, cardiovascular, and other tissues, and 3) a safe, early, cost effective, time efficient, low power, small mass and small volume alternative to artificial gravity by centrifugation for many physiologic systems.<br><br>Our finding of the magnitude and mechanism of force production by LBNP has important implications for simulating gravity in space and increasing weightbearing on Earth without the use of a centrifuge. Our upright-standing results indicate that ground reaction force increases by approximately one equivalent body weight for each 50 mm Hg LBNP. Our supine data suggest that during microgravity, uniaxial loading of lower-body tissues can be comfortably produced by a similar level of LBNP. The use of a different air pressure separating the upper and lower body, such as proposed in this project, distributes the net force uniformly over the entire upper surface of the body. This concept thereby avoids the discomfort of localized high pressures typical of bungee cord harness systems. Also, a broad, flexible waist seal with shoulder straps increases the effective surface area through which the pressure differential creates force, applying an additional force through the shoulders and spine to the lower extremities. This latter configuration reduces the negative pressure necessary to generate a given force at the feet and normalizes the musculoskeletal and cardiovascular stresses to those of upright activity on Earth. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Upright LBNP exercise on Earth may improve performance of athletes (Macias B, H Tran Cao, B Lee, M Bawa, E Groppo, R Pedowitz, A Hargens. Physiologic responses of lower body negative pressure treadmill exercise as a training modality. American College of Sports Medicine SW Chapter, 2002 Student Award; Macias BR, ER Groppo, M Bawa, HS Tran Cao, B Lee, RA Pedowitz and AR Hargens. Lower body negative pressure treadmill exercise is more comfortable and produces similar physiological responses as weighted vest exercise. International Journal of Sports Medicine 27:1-5, 2006). Conversely, lower body positive pressure (LBPP) decreases weight bearing and edema and thus, aids rehabilitation in orthopaedic patients (Cutuk A, ER Groppo, EJ Quigley, K White, RA Pedowitz and AR Hargens. Ambulation in simulated fractional gravity using lower body positive pressure: cardiovascular safety and gait analyses. Journal of Applied Physiology 101:771-777, 2006). This concept of post-surgical unweighting exercise has advantages over other rehabilitation modalities (Eastlack RK, AR Hargens, ER Groppo, GC Steinbach, KK White and RA Pedowitz. Lower body positive-pressure exercise after knee surgery. Clinical Orthopaedic and Related Research 431:213-219, 2005: Recipient of the Jacquelin Perry Award from the Orthopaedic Rehabilitation Association; Macias BR, ER Groppo, RK Eastlack, DE Watenpaugh, SM Lee, SM Schneider, WL Boda, SM Smith, A Cutuk, RA Pedowitz, RS Meyer, and AR Hargens. Space exercise and Earth benefits. Current Pharmaceutical Biotechnology 6:305-317, 2005) and may also aid rehabilitation of crew members after prolonged microgravity. |
| | New Results from WISE-2005<br>(A. Hargens, NASA Grant NNJ04HF71G to UCSD)<br><br>1) PeakVO2 maintained in Exercise Group despite reduced frequency (3-4 LBNPex/week compared to 6/week in an earlier bed rest study of identical twins at UCSD).<br><br>2) Bed rest subjects maintained peakVO2 for at least 5 days following last LBNPex session.<br><br>3) Heart mass was decreased in Control Group, unchanged in Nutrition Group, and increased in Exercise Group.<br><br>4) Heart mass loss is similar in females and males and continues to decrease with bed-rest duration.<br><br>5) Orthostatic tolerance was maintained better in Exercise Group than Control and Nutrition Groups, despite not having |

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| **Task Progress:** | LBNPex for 72 hours prior to test.<br><br>6) In agreement with flight results, women have lower orthostatic tolerance than men.<br><br>7) Muscle strength maintained in Exercise Group, but not in Control and Nutrition Groups. 8) Combined resistive and aerobic (3-4 LBNPex per week) didn't protect bone in the 60 day WISE bed rest as well as LBNPex alone (6 LBNPex per week in the UCSD 30 day bed-rest study of identical twins).<br><br>9) WISE-2005 allowed a very comprehensive study of a combined resistive and aerobic exercise countermeasure during 60 days bed rest.<br><br>10) WISE-2005 fostered a comprehensive and productive collaboration between investigators from NASA, ESA, CNES and CSA which may set an example for future international ground-based and flight collaborations. |
| **Bibliography Type:** | Description: (Last Updated: 08/25/2025) |
| **Abstracts for Journals and Proceedings** | Macias BR, Schneider SM, Lee SMC, Guinet P, Meuche S, Hughson RL, Smith SM, Zwart SE, Watenpaugh DE, Hargens AR. "WISE-2005: Lower Body Negative Pressure Treadmill Exercise and Resistive Exercise Countermeasure Maintain Physiologic Function in Women during 60-days of Bed Rest." 23rd Annual Meeting of the American Society for Gravitational and Space Biology, Moffett Field, CA, October 2007.<br>Gravitational and Space Biology 2007 Oct;21(1):17. , Oct-2007 |
| **Articles in Peer-reviewed Journals** | Sayson JV, Hargens AR. "Pathophysiology of low back pain during exposure to microgravity." Aviat Space Environ Med. 2008 Apr;79(4):365-73. Review.<br>http://www.ingentaconnect.com/content/asma/asem/2008/00000079/00000004/art00001 , Apr-2008 |
| **Articles in Peer-reviewed Journals** | Holt JA, Macias BR, Schneider SM, Watenpaugh DE, Lee SM, Chang DG, Hargens AR. " WISE 2005: Aerobic and resistive countermeasures prevent paraspinal muscle deconditioning during 60-days bed rest in women." J Appl Physiol (1985). 2016 Feb [Epub ahead of print] http://dx.doi.org/10.1152/japplphysiol.00532.2015 ; PubMed PMID: 26893030 , Feb-2016 |
| **Articles in Peer-reviewed Journals** | Macaulay TR, Macias BR, Lee SMC, Boda WL, Watenpaugh DE, Hargens AR. "Treadmill exercise within lower-body negative pressure attenuates simulated spaceflight-induced reductions of balance abilities in men but not women." npj Microgravity. 2016;2:16022. Published online: 30 June 2016. http://dx.doi.org/10.1038/npjmgrav.2016.22 , Jun-2016 |
| **Articles in Peer-reviewed Journals** | Schneider SM, Lee SM, Feiveson AH, Watenpaugh DE, Macias BR, Hargens AR. "Treadmill exercise within lower body negative pressure protects leg lean tissue mass and extensor strength and endurance during bed rest." Physiol Rep. 2016 Aug;4(15). http://dx.doi.org/10.14814/phy2.12892 ; PubMed PMID: 27495299; PubMed Central PMCID: PMC4985554 , Aug-2016 |
| **Articles in Peer-reviewed Journals** | Hargens AR, Kim JM, Cao P. "Accuracy of water displacement hand volumetry using an ethanol and water mixture." Aviat Space Environ Med. 2014 Feb;85(2):187-90. https://doi.org/10.3357/ASEM.3485.2014 ; PMID: 24597165 , Feb-2014 |
| **Articles in Peer-reviewed Journals** | Smith SM, Zwart SR, Heer M, Lee SM, Baecker N, Meuche S, Macias BR, Shackelford LC, Schneider S, Hargens AR. "WISE-2005: Supine treadmill exercise within lower body negative pressure and flywheel resistive exercise as a countermeasure to bed rest-induced bone loss in women during 60-day simulated microgravity." Bone. 2008 Mar;42(3):572-81. PMID: 18249055 , Mar-2008 |
| **Articles in Peer-reviewed Journals** | Dorfman TA, Levine BD, Tillery T, Peshock RM, Hastings JL, Schneider SM, Macias BR, Biolo G, Hargens AR. "Cardiac atrophy in women following bed rest." J Appl Physiol. 2007 Jul;103(1):8-16. Epub 2007 Mar 22. PMID: 17379748 , Jul-2007 |
| **Articles in Peer-reviewed Journals** | Macias BR, Groppo ER, Bawa M, Tran Cao HS, Lee B, Pedowitz RA, Hargens AR. "Lower body negative pressure treadmill exercise is more comfortable and produces similar physiological responses as weighted vest exercise." Int J Sports Med. 2007 Jun;28(6):501-5. Epub 2006 Nov 16. PMID: 17111308 , Jun-2007 |
| **Articles in Peer-reviewed Journals** | Macias BR, Groppo ER, Eastlack RK, Watenpaugh DE, Lee SM, Schneider SM, Boda WL, Smith SM, Cutuk A, Pedowitz RA, Meyer RS, Hargens AR. "Space exercise and Earth benefits." Curr Pharm Biotechnol. 2005 Aug;6(4):305-17. Review. PMID: 16101469 , Aug-2005 |
| **Awards** | Macias, Brandon R. "Tom K. Scott Award for best abstract and oral presentation in the field of Gravitational and Space Biology, American Society for Gravitational and Space Biology Annual Meeting, 27 October 2007." Oct-2007 |
| **Awards** | Macias BR, Schneider SM, Lee SMC, Guinet P, Meuche S, Hughson RL, Smith SM, Zwart SE, Watenpaugh DE, Hargens AR. "First Place Poster Award for “WISE-2005: Lower Body Negative Pressure Treadmill Exercise and Resistive Exercise Countermeasure Maintain Physiologic Function in Women during 60-days of Bed Rest.“ American Society for Gravitational and Space Biology annual meeting, October 2007. Poster B10." Oct-2007 |
| **Awards** | Hargens AR. "Hallman Lecture Award for Distinguished Visiting Professor, University of Waterloo, Canada, October 2006." Oct-2006 |
| **Awards** | Meuche S, Schneider SM, Lee SMC, Macias BR, Watenpaugh DE, Hargens AR. "Student Award from American College of Sports Medicine, November 2006." Nov-2006 |
| **Books/Book Chapters** | Hargens AR, Petterson K, Millard RW. "Giraffe Cardiovascular Adaptations to Gravity. " in "Endothelial biomedicine : a comprehensive reference." Ed. W.C. Aird. Cambridge ; New York : Cambridge University Press, 2007. pp. 94-101., Mar-2007 |
| **Significant Media Coverage** | Hargens AR. "Spaceflight Deconditioning and Aging: What We've Learned in Bed." National TV and public lecture for the UCSD Stein Institute for Research on Aging, January 2007., Jan-2007 |